Shawaz Imam

Software Engineer — Backend · Platform · Applied AI

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SUMMARY

Full-stack engineer with a backend and systems focus. I'm the sole engineer on a production multi-tenant SaaS (TypeScript, PostgreSQL, real-time AI voice) serving a property-management customer, and I build performance-critical services in Rust. I own the whole stack: schema, APIs, UI, CI/CD, infrastructure, and operations. I work test-first and rely on real-system verification, not mocks alone.

SKILLS

Languages	TypeScript, Rust, Python, SQL, JavaScript, Solidity, Bash, Nix
Backend	NestJS, Node.js, Fastify, Axum, Actix-web, tokio, Django/DRF, FastAPI; REST, WebSockets, webhooks
Frontend	React, Next.js, React Native/Expo, Tailwind, Redux; Playwright; i18n/RTL
Data	PostgreSQL (row-level security, migrations), Drizzle, SQLx, Prisma, Redis/BullMQ, SQLite, Qdrant
AI & voice	LLM tool use and fallback chains, MCP servers, RAG, Pipecat, streaming STT/TTS, Ollama, agentic dev workflows
Infra	GCP, Terraform, Docker, GitHub Actions, NixOS, systemd, Cloudflare, Firebase, Linux

EXPERIENCE

Software Engineer & Facilities Manager

Dec 2024 – Present

Property management company, Greater Boston

- Designed, built, and operate, as sole engineer, a **multi-tenant property-maintenance SaaS** with an AI front desk. It triages requests from phone, SMS, email, and web, opens work orders, dispatches vendors, and syncs to the customer's property-management system. TypeScript monorepo with 11 apps, 37 packages, 122 migrations, and about 1,900 commits.
- Enforced tenant isolation in the database: **forced Postgres row-level security** across all org-scoped tables, a least-privilege runtime role, and transaction-scoped context that prevents cross-org leaks through pooled connections. Verified with cross-org attack tests.
- Built a **real-time AI voice agent** (carrier media streams → streaming STT → LLM → TTS) with multi-vendor fallback, plus a synthetic-caller harness that places real calls in automated tests. The harness found stacked defects that had caused dropped calls.
- Implemented **safety-critical triage** (life-safety and seasonal no-heat floors that config can't lower) and an emergency dispatch cascade that pages on-call staff when every vendor has been tried.
- Integrated **Stripe subscriptions**: signed webhooks, protection against out-of-order events, and idempotency keys hashed from request parameters. A live sandbox contract suite (50 tests) caught three defects that mocked unit tests had passed.
- Built CI/CD: an 8-job GitHub Actions pipeline (secret scan, smoke boot, visual and accessibility checks, end-to-end tests) and a one-command deploy with rollback behind a Playwright gate. Found and fixed a critical auth-bypass CVE in a dependency, a SQL-injection advisory, and a token-rotation race.
- Built a **guarded LLM agent** (MCP server) for operational records. Writes follow propose → confirm → execute with single-use, payload-bound tokens, and the agent runs in a sandboxed service limited to one outbound host.

- Manage facilities for 25 properties, including 24/7 emergency response and vendor coordination. That daily contact with operations drives the product's requirements.

TypeScript · NestJS · Next.js · Expo · PostgreSQL · Drizzle · Redis/BullMQ · Python · Pipecat · Twilio/Telnyx · Stripe · Terraform · GCP · Cloudflare

Independent Software Engineer 2024 – Present

Rust systems, DeFi, and applied AI (personal projects)

- **Multi-chain liquidation engine** (Rust, about 21K lines, 320+ tests): an event-driven pipeline that simulates each trade in a local EVM (revm) before any RPC call. Includes a 15-state opportunity lifecycle, circuit breakers, Multicall3 batching, and a Solidity flash-loan contract tested with Foundry and deployed on 5 EVM chains.
- **B2B marketplace backend** (Rust/Axum/SQLx, 5 crates, 150+ tests including Testcontainers): separate Postgres schemas and a backend that is the only writer keep supplier identity hidden from buyers. Append-only money and audit records, Stripe deposits, TOTP MFA.
- **Infrastructure**: a 6-host NixOS fleet managed from one flake, with VM tests and CI, local GPU inference, a RAG index over my codebases, and fully local voice assistants.

EDUCATION

B.A., Interdisciplinary Studies (Applied Psychology), University of Massachusetts Amherst 2024

Full Stack Developer Certification, Clarusway 2023